

CHAPTER 21

NUCLEAR ATOM

5.1 The nuclear model of the atom

5.1.1 The atom

- 1 Describe the structure of the atom in terms of a positively charged nucleus and negatively charged electrons in orbit around the nucleus
- 2 Describe how alpha-particle scattering experiments provide evidence for:
 - (a) a very small nucleus surrounded by mostly empty space
 - (b) a nucleus containing most of the mass of the atom
 - (c) a nucleus that is positively charged

5.1.2 The nucleus

- 1 Describe the composition of the nucleus in terms of protons and neutrons
- 2 Describe how atoms form positive ions by losing electrons or negative ions by gaining electrons
- 3 Define the terms proton number (atomic number) Z and nucleon number (mass number) A and be able to calculate the number of neutrons in a nucleus
- 4 Explain the term nuclide and use the nuclide notation A_ZX
- 5 Explain what is meant by an isotope and state that an element may have more than one isotope

1. O/N/23/11/Q35

Which particles are in the nucleus of a large atom?

- A** neutrons and electrons only
- B** protons and electrons only
- C** protons and neutrons only
- D** protons, neutrons and electrons

2. M/J/23/12/Q33

The results of the alpha-particle scattering experiment give evidence for which of the following?

- A** nuclear fusion
- B** radioactive decay
- C** the existence of isotopes
- D** the size of the nucleus

3. M/J/23/12/Q34

A nucleus of an isotope of cobalt is represented by the symbol $^{59}_{27}\text{Co}$.

What is the composition of this nucleus?

	number of protons	number of neutrons
A	27	32
B	27	59
C	59	27
D	59	32

4. M/J/23/12/Q35

The only stable isotope of gold, Au, has 79 protons and 118 neutrons in a nucleus of an atom.

How many electrons are there in a neutral atom of this isotope?

- A** 39 **B** 79 **C** 118 **D** 197

5. M/J/23/11/Q35

Atoms P and Q are isotopes of the same element.

How does the composition of an atom of P compare with the composition of an atom of Q?

	number of protons	number of neutrons
A	different	different
B	different	same
C	same	different
D	same	same

6. M/J/23/11/Q36

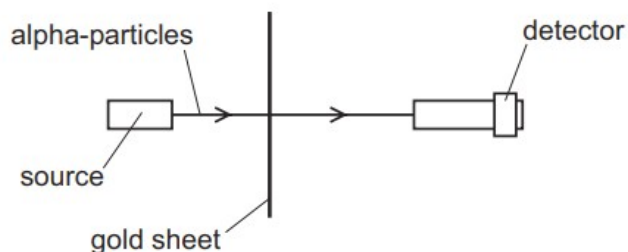
The nucleus of an atom emits an alpha-particle.

How do the proton number and nucleon number change?

	proton number	nucleon number
A	decreases by 2	decreases by 4
B	decreases by 4	decreases by 2
C	increases by 2	increases by 4
D	increases by 4	increases by 2

7. M/J/22/11/Q40

In the Geiger-Marsden experiment, alpha-particles are fired at a thin gold sheet.



Most alpha-particles pass straight through the thin gold sheet.

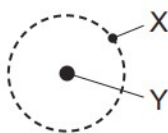
A few are deflected.

What can be deduced from this experiment?

- A** The nucleus is very small.
- B** The nucleus has no charge.
- C** Electrons surround the nucleus.
- D** Electrons have a negative charge.

8. O/N/21/11/Q40

In the simple model of an atom, X orbits around Y.



What are X and Y?

	X	Y
A	electron	nucleus
B	neutron	electron
C	nucleus	proton
D	proton	neutron

9. O/N/21/12/Q40

How do the proton numbers (atomic numbers) and the nucleon numbers (mass numbers) of two different isotopes of the same element compare with each other?

	proton number	nucleon number
A	different	different
B	different	same
C	same	different
D	same	same

10. M/J/21/11/Q40

Which particles are found inside the nucleus of an atom?

- A** neutrons and electrons
- B** electrons and protons
- C** neutrons only
- D** neutrons and protons

11. O/N/20/12/Q40

Which row shows the atomic structure of a neutral atom with a nucleon number of 9?

	number of neutrons	number of protons	number of electrons
A	4	5	4
B	4	5	5
C	5	4	5
D	5	5	4

12. M/J/20/11/Q37

The nuclide notation for an isotope of X is ${}^A_Z\text{X}$.

How many neutrons are there in a nucleus?

- A** A **B** $A - Z$ **C** $Z - A$ **D** Z

13. M/J/19/11/Q40

A neutral atom of boron contains 5 electrons and 5 protons.

The nucleon number (mass number) of the atom is 11.

How many neutrons are there in the atom?

- A** 1 **B** 5 **C** 6 **D** 11

14. O/N/18/11/Q38

The results of the alpha-particle scattering experiment gave evidence for which of the following?

- A** nuclear fusion
B radioactive decay
C the existence of isotopes
D the nuclear atom

15. O/N/18/11/Q40

One nucleus of a uranium isotope contains 238 nucleons of which 146 are neutrons.

Which is the nucleus of another isotope of uranium?

	number of nucleons in the nucleus	number of neutrons in the nucleus
A	235	143
B	237	147
C	238	144
D	241	146

16. O/N/18/12/Q40

Which statement about two different isotopes of the same element is correct?

- A** Both isotopes have the same number of electrons in a nucleus.
B Both isotopes have the same number of protons in a nucleus.
C Both isotopes have the same number of neutrons in a nucleus.
D Both isotopes have the same number of nucleons in a nucleus.

17. M/J/18/12/Q40

How many nucleons are in one neutral atom of the krypton isotope $^{84}_{36}\text{Kr}$?

- A** 36 **B** 48 **C** 84 **D** 120

18. M/J/18/11/Q40

A radioactive nucleus is represented as $^{230}_{92}\text{U}$.

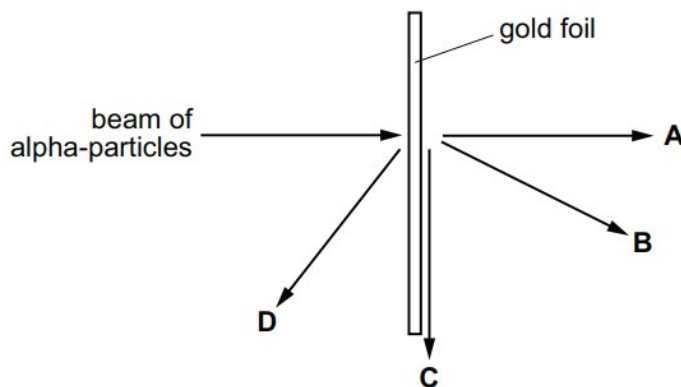
Which row represents another isotope of the same element?

	number of protons in nucleus	number of neutrons in nucleus
A	90	140
B	91	138
C	92	139
D	93	136

19. O/N/17/11/Q37

A narrow beam of alpha-particles is fired at a thin piece of gold foil.

What is the final direction of the largest number of alpha-particles?

**20. O/N17/11/Q38**

A nucleus of phosphorus $^{32}_{15}\text{P}$ emits a beta-particle to form a new nucleus.

What is the nucleon number and what is the proton number of the new nucleus?

	nucleon number (mass number)	proton number (atomic number)
A	28	13
B	31	14
C	31	15
D	32	16

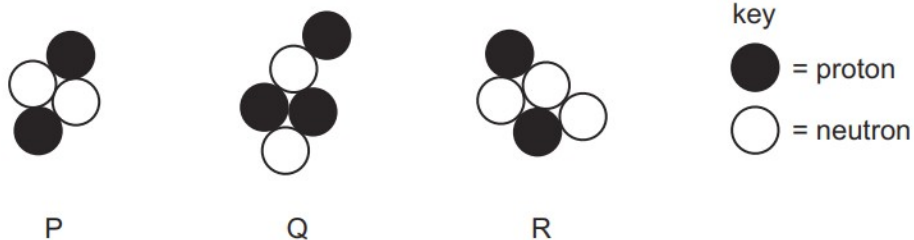
21. O/N/17/11/Q39

How does a nucleus of the isotope chlorine-37 differ from a nucleus of the isotope chlorine-35?

- A It contains one more neutron and one more electron.
- B It contains one more proton and one more electron.
- C It contains two more neutrons.
- D It contains two more protons.

22. M/J/16/12/Q40

The diagrams represent the nuclei of three atoms.



Which are isotopes of the same element?

- A P and Q only
- B P and R only
- C Q and R only
- D P, Q and R

23. M/J/16/11/Q40

Which statement about a nucleus of $^{15}_7\text{N}$ is correct?

- A The nucleus contains 7 neutrons and 8 protons.
- B The nucleus contains 7 neutrons and 15 protons.
- C The nucleus contains 7 protons and 8 neutrons.
- D The nucleus contains 7 protons and 15 neutrons.

24. O/N/15/12/Q40

In the Geiger-Marsden experiment, a beam of alpha-particles is fired at a very thin sheet of gold foil, in a vacuum.

What is deduced from this experiment?

- A Alpha-particles are repelled by electrons.
- B Atoms contain air.
- C Electrons are found in atomic nuclei.
- D Nuclei are much smaller than atoms.

25. O/N/14/11/Q40

How many nucleons are in one neutral atom of the krypton isotope $^{84}_{36}\text{Kr}$?

- A** 36 **B** 48 **C** 84 **D** 120

26. O/N/14/12/Q40

A neutral atom of chlorine-37 is compared with a neutral atom of argon-37.

How do the number of electrons and the number of neutrons in each of the atoms compare?

	number of electrons	number of neutrons
A	different	different
B	different	same
C	same	same
D	same	different

27. O/N/13/11/Q40

The compositions of four nuclei are shown in the table.

nucleus	number of protons	number of neutrons	number of nucleons
P	88	141	229
Q	88	136	224
R	89	139	228
S	92	136	228

Which two nuclei are isotopes of the same element?

- A** P and Q **B** P and S **C** Q and S **D** R and S

28. O/N/13/12/Q40

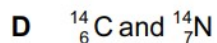
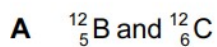
A nucleus of phosphorus $^{32}_{15}\text{P}$ emits a beta-particle to form a new nucleus.

What is the nucleon number and what is the proton number of the new nucleus?

	nucleon number (mass number)	proton number (atomic number)
A	28	13
B	31	14
C	31	15
D	32	16

29. M/J/13/11/Q40

Which nuclides have the same number of neutrons in a nucleus?

**30. O/N/12/12/Q38**

How many protons are in the nucleus of an atom of radium, $^{226}_{88}\text{Ra}$?

A 88

B 138

C 226

D 314

31. M/J/12/12/Q38

The nucleus of a helium atom is represented as ^4_2He .

What does a neutral atom of helium contain?

	electrons	protons	neutrons
A	2	2	2
B	2	4	2
C	4	2	2
D	4	4	2

32. O/N/11/11/Q38

One isotope of carbon is $^{14}_6\text{C}$.

How many neutrons and protons does each atom of this isotope contain?

	number of neutrons	number of protons
A	6	6
B	6	8
C	8	6
D	14	6

33. M/J/11/12/Q39

Chlorine exists as two isotopes. One has a nucleon number (mass number) of 35 and the other has a nucleon number (mass number) of 37.

Which table shows the correct numbers of protons and neutrons in the isotopes?

A

	number of protons	number of neutrons
isotope 1	17	18
isotope 2	17	20

B

	number of protons	number of neutrons
isotope 1	18	17
isotope 2	20	17

C

	number of protons	number of neutrons
isotope 1	35	17
isotope 2	37	17

D

	number of protons	number of neutrons
isotope 1	17	35
isotope 2	17	37

34. O/N/10/11/Q39

$^{15}_7\text{N}$ is a nuclide of nitrogen.

How many electrons are there in a neutral atom of $^{15}_7\text{N}$?

A 7**B** 8**C** 15**D** 22**35. M/J/10/12/Q39**

Proton number is another name for atomic number. Nucleon number is another name for mass number.

What are isotopes?

A nuclei with different proton numbers and different nucleon numbers**B** nuclei with different proton numbers and the same nucleon number**C** nuclei with the same proton number and different nucleon numbers**D** nuclei with the same proton number and the same nucleon number**36. O/N/09/Q40**

A nuclide of strontium is represented by the symbol $^{88}_{38}\text{Sr}$.

What does the nucleus contain?

A 38 electrons and 50 neutrons**B** 38 neutrons and 38 protons**C** 38 neutrons and 50 protons**D** 38 protons and 50 neutrons

37. M/J/09/Q40

What are the numbers of neutrons, protons and electrons in a neutral atom of $^{235}_{92}\text{U}$?

	number of neutrons	number of protons	number of electrons
A	92	143	143
B	92	235	235
C	143	92	92
D	235	92	92

38. O/N/08/Q39

The radioactive nucleus $^{87}_{37}\text{Rb}$ decays with the emission of a beta-particle.

How many protons and neutrons are left in the nucleus?

	protons	neutrons
A	37	49
B	37	50
C	38	49
D	38	87

39. O/N/07/Q40

The neutral atoms of all isotopes of the same element contain the same number of

- A** electrons and protons.
- B** electrons and neutrons.
- C** neutrons only.
- D** neutrons and protons.

40. M/J/07/Q40

A nuclide has the notation $^{48}_{23}\text{X}$.

Which line in the table describes a different isotope of this nuclide?

	proton number (atomic number)	nucleon number (mass number)
A	23	50
B	24	48
C	48	24
D	50	23

41. O/N/06/Q37

A nucleus of sodium, Na, has 11 protons and 12 neutrons.

Which symbol represents this nucleus?

- A** ${}^{11}_{12}\text{Na}$ **B** ${}^{12}_{11}\text{Na}$ **C** ${}^{23}_{11}\text{Na}$ **D** ${}^{23}_{12}\text{Na}$

42. M/J/06/Q40

Which conclusion can be drawn from the Geiger-Marsden alpha-particle scattering experiment?

- A** A positive charge is spread throughout the atom.
B Electrons are arranged in orbits.
C Electrons are negatively charged.
D There is a dense nucleus in the atom.

43. O/N/05/Q39

A nuclide of the element plutonium is ${}^{242}_{94}\text{Pu}$.

What is the number of neutrons in its nucleus?

- A** 94 **B** 148 **C** 242 **D** 336

44. O/N/05/Q40

Which statement defines isotopes?

- A** two (or more) nuclides which have the same number of protons but different numbers of electrons
B two (or more) nuclides which have the same number of neutrons but different numbers of electrons
C two (or more) nuclides which have the same number of neutrons but different numbers of protons
D two (or more) nuclides which have the same number of protons but different numbers of neutrons

45. M/J/05/Q40

Between 1909 and 1911, Geiger and Marsden carried out experiments in which alpha particles were fired at metal foil. Most of the alpha particles passed through the foil with small deflections, but some were deflected through a large angle.

These results suggest that

- A** atoms contain clouds of electrons through which some alpha particles cannot pass.
B atoms contain neutrons that alpha particles bounce off.
C atoms have positive and negative charges spread throughout their volume.
D atoms have positive charges concentrated in a small volume.

46. O/N/04/Q40

Deuterium ${}^2_1\text{H}$ and tritium ${}^3_1\text{H}$ are two isotopes of hydrogen.

Compared to a deuterium atom, how many protons and neutrons does a tritium atom have?

	protons	neutrons
A	more	more
B	more	same
C	same	more
D	same	same

ANSWERS

1. C
2. D
3. A
4. B
5. C
6. A
7. A
8. A
9. C
10. D
11. B
12. B
13. C
14. D
15. A
16. B
17. C
18. C
19. A
20. D
21. C
22. B
23. C
24. D
25. C
26. A
27. A
28. D
29. C
30. A
31. A
32. C
33. A
34. A
35. C
36. D
37. C
38. C
39. A
40. A
41. C
42. D
43. B
44. D
45. D
46. C

1. O/N/22/21/Q5

Fig. 5.1 is a diagram that shows the number of particles in a charged atom (ion) of beryllium (Be).

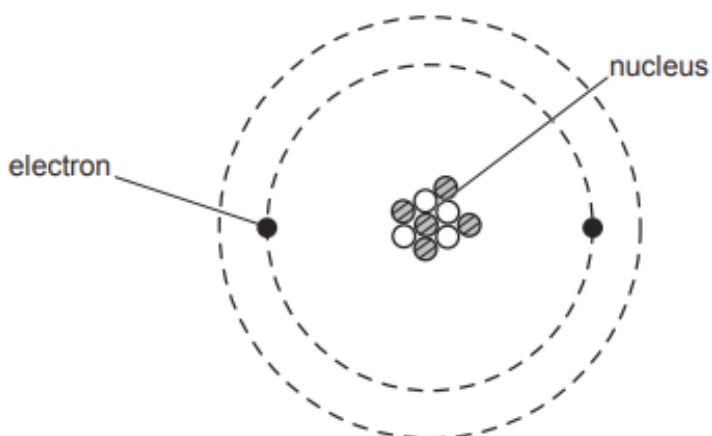


Fig. 5.1

- (a) The charge on an electron is $-1.6 \times 10^{-19} \text{ C}$. The charge on the ion is $+3.2 \times 10^{-19} \text{ C}$.

(i) Deduce, from the charge on the ion, the number of protons in its nucleus.

Show your working.

number of protons = [2]

(ii) Write down, in nuclide notation, the symbol for the nucleus of this charged atom.

..... [1]

- (b) The isotope in Fig. 5.1 is the only stable isotope of beryllium.

Explain the term 'isotope'.

.....

 [2]

- (c) Explain what prevents electrons in an atom from escaping.

.....

 [2]

[Total: 7]

2. M/J/15/22/Q8

Two isotopes of hydrogen are written as ${}^1_1\text{H}$ and ${}^2_1\text{H}$.

- (a) Complete Fig. 8.1 to show the number of protons and neutrons in one nucleus of each of these isotopes of hydrogen.

	number of protons	number of neutrons
${}^1_1\text{H}$		
${}^2_1\text{H}$		

Fig. 8.1

[2]

- (b) Explain, using ideas about electrons, neutrons and protons, why atoms of ${}^1_1\text{H}$ and ${}^2_1\text{H}$ are uncharged.

.....

.....

.....

.....

.....

.....

..... [3]

3. O/N/12/22/Q8

The nuclei of four neutral atoms are represented using nuclide notation as:



- (a) Neutral atoms also contain electrons. State which of the four nuclei is surrounded by the greatest number of electrons.

..... [1]

- (b) From these four nuclei, state the **two** nuclei that have the same number of

- (i) protons,

..... [1]

- (ii) nucleons,

..... [1]

- (iii) neutrons.

..... [1]

4. O/N/09/Q8

(a) State the number of

(i) electrons in a neutral atom of the isotope ${}^7_3\text{Li}$,

number of electrons = [1]

(ii) nucleons in an atom of the isotope ${}^{208}_{82}\text{Pb}$,

number of nucleons = [1]

(iii) protons in an atom of the isotope ${}^{23}_{11}\text{Na}$.

number of protons = [1]

(b) An atom of the isotope chlorine-35 contains 17 protons.

For an atom of the isotope chlorine-37, state the number of

(i) protons,

number of protons = [1]

(ii) neutrons.

number of neutrons = [1]

ANSWERS

1.

Question	Answer	Marks
5(a)(i)	3.2×10^{-19} / (-1.6×10^{-19}) or 2 more protons than electrons shown by charges	B1
	$(2 + 2 =) 4$	B1
5(a)(ii)	${}^9_4\text{Be}$	B1
5(b)	a form of an element	B1
	with a particular number of neutrons / nucleons	B1
5(c)	electrons and nucleus opposite charged or electrons negative and nucleus positive	B1
	(electrostatic / electric) force of attraction or unlike charges attract	B1

2.

- (a) left column both 1 B1
right column 0 **and** 1 B1
- (b) (at least one of the atoms) contain same number of electrons and protons B1
or have 1 electron and 1 proton
charge on electron and proton opposite B1
or electron negative **and** proton positive
or charge on electron neutralises/cancels/balances proton charge
neutrons have no charge B1

3.

- (a) ${}^{15}_8\text{O}$ / oxygen-15 / oxygen (nucleus) B1
- (b) (i) ${}^{12}_6\text{C}$ **and** ${}^{14}_6\text{C}$ / carbon-12 **and** carbon-14 / the two carbon nuclei B1
- (ii) ${}^{14}_6\text{C}$ **and** ${}^{14}_7\text{N}$ / carbon-14 **and** nitrogen-14 B1
- (iii) ${}^{14}_7\text{N}$ **and** ${}^{15}_8\text{O}$ / nitrogen-14 **and** oxygen-14 / the nitrogen and oxygen nuclei B1

4.

- (a) (i) 3 **cao** B1
- (ii) 208 **cao** B1
- (iii) 11 **cao** B1
- (b) (i) 17 **cao** B1
- (ii) 20 **cao** B1